



MULTITECH 

CONNECT

2024 Partner Conference



Solar Power for LoRaWAN Gateways

AGENDA

- Introduction to Voltaic
- Design perspective on solar systems
- Running the Multitech IP67 Gateway
- Properly sizing solar systems
- Product roadmap



BACKGROUND

We design and manufacture the solar panels and solar power systems for Internet of Things (IoT) devices.

We are based in Brooklyn, NY.

We work with companies to power industrial and IoT devices. They include OEMs, product design firms, universities and NGOs. They include **market leaders** in:

- ☞ GPS asset tracking
- ☞ Agricultural technology
- ☞ Marine applications
- ☞ Environmental and weather sensing
- ☞ Communications equipment
- ☞ Parking, lighting and smart city applications
- ☞ Smart cameras

Voltaic will ship roughly 1 million solar panels in 2024.



DESIGN PERSPECTIVE FOR GATEWAYS AND LARGER SYSTEMS

We deployed a solar-powered **LoRaWAN gateway** for FloodNet in the Brooklyn Navy Yard as a proof of concept. Goal was to provide a backup in case their mains powered gateways shut off due to power outages.

Features:

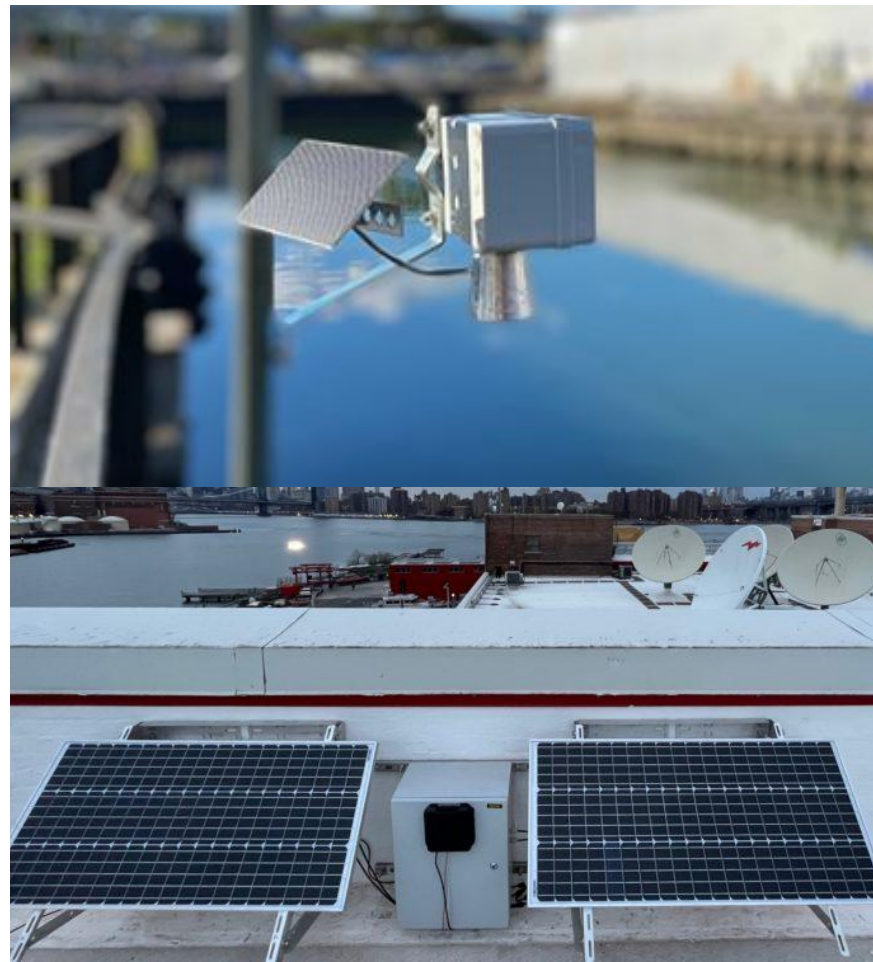
- ☞ 4.5W continuous or 108Wh per day
- ☞ 1,440Wh lead-acid battery
- ☞ 2 x 100W 18V solar panels

Challenges:

- ☞ Siting permission on roof (6 months)
- ☞ Part sourcing - nuts, glands, brackets, etc.
- ☞ Lead acid battery very heavy in a bulky enclosure
- ☞ Long installation time - panel brackets in particular

Upside:

Battery monitoring application allowed us to track system performance



CORE SOLAR SYSTEMS DESIGN CHOICES

Prioritized **product lifetime**, **deployment speed**, and **battery health monitoring**

1. **Lithium Iron Phosphate battery** for power density and charge cycles
2. **Eliminate battery enclosure** to reduce system size
3. Panel **bracket pre-assembled**, requires only 6 bolts
4. Monitoring capability integrated into battery - **effectively a node**
5. Battery **attaches cleanly** to bracket
6. Mix and match panels and battery sizes **depending on load and location**
7. PoE option integrated into battery



MULTITECH IP67 GATEWAY

Currently running in upstate New York from PoE.
5.6 Watts x 24h = 134 Watt hours per day

D1003 - Voltaic | Multitech IP67 | 12V100AH

V103

View Edit

Battery Output Status
Since last transmission

Online

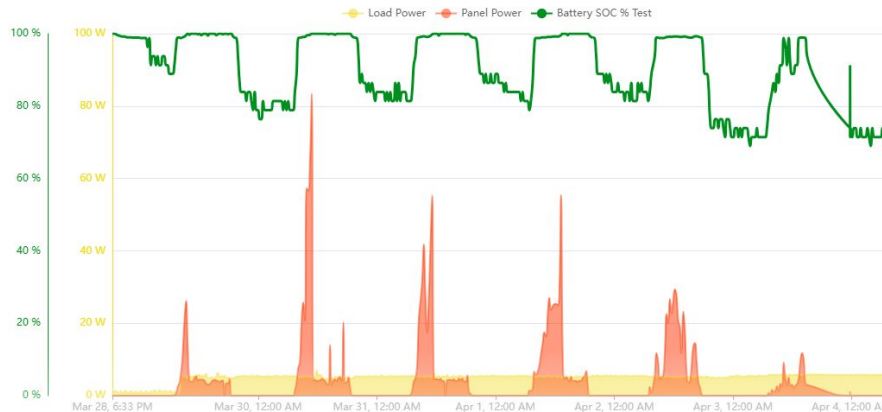
Battery State of Charge
Last measured value

99%

Battery Temperature
Last measured value

-1.00 °C

Power In, Power Out, and Battery State of Charge (SOC)
Last 7 days



Available Power from Sun > **1.5x** Power Consumed Per Day

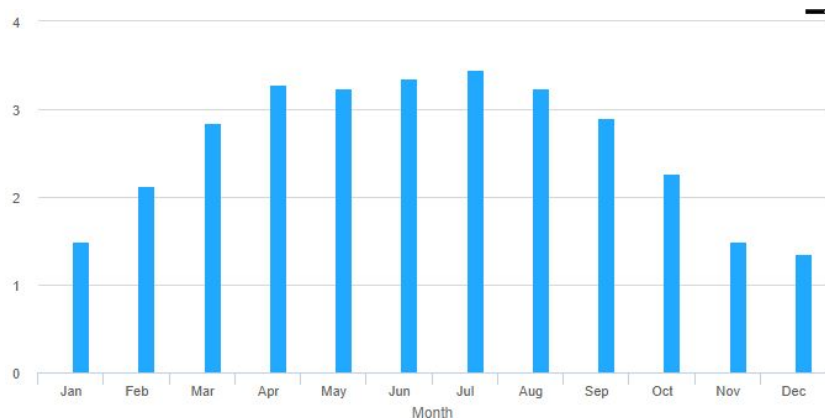
Battery Runtime with no Sun > **4 days**

SYSTEM SIZING - AVAILABLE POWER

We look at the “equivalent sun” a fixed panel will receive over an average day.

We size a system based on the worst performing time of year.

Hours of Sun Per Day by Month in Munich*



*From PVGIS - Photovoltaic Geographic Information System

Location	Hours of Sun in December
Munich	1.35
Malaga	3.04
London	1.07
Stockholm	0.44

SYSTEM SIZING - IP67 GATEWAY

The IP67 Gateway uses 5.6 Watts including our PoE injector or **134 Watt hours per day**

Panels should produce 1.5X expected power consumption or > than **200 Watt hours per day**

Battery provides buffer for bad weather and we recommend > **45Ah** for 4 days backup

Solar Panel Sizing for December

Location	Hours of Sun in December	Power Produced 100 Watt Solar Panel	Power Produced 200 Watt Solar Panel
Munich	1.35	135Wh	270Wh
Malaga	3.04	304Wh	608Wh
London	1.07	107Wh	214Wh
Stockholm	0.44	44Wh	88Wh

Battery Sizing

Battery	18Ah	60Ah	100Ah
Days Runtime	~1.6	~5	~9

Sizing tool on website

— OPTIONS IF APPLICATION IS UNDERPOWERED —

1. **Increase system size** - this adds cost, size and weight
2. **Decrease power consumption** - developing an option to turn off load on a schedule, potentially only during worst months



Application Opportunities

1. Available for sizing discussion at jeff@voltaicsystems.com
2. Datasheets for review available

Product

1. Addition of email alerts from monitoring platform and predictive analytics
2. Load shutoff option
3. Continued minor enhancements to improve ease of deployment
4. Addition of low temperature battery option
5. Improving distribution in EU

SYSTEM SIZING - COLD TEMPERATURE IMPACT

We currently do not recommend our systems for locations with high frequency of days where high temperature is below 0C.

Issues:

- ⌚ We stop charging below 0C
- ⌚ Charge capacity drops in cold temperature

Options for cold temperature:

- ⌚ Increase battery and panel size
- ⌚ Use alternate chemistry like lead acid

Improvements:

- ⌚ Voltaic producing pilot of modified chemistry LFP that can charge to -20C.